87/100

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Duration: 60 Minutes** **Total Marks: 25**

Question Count: 5

Code of Conduct

I Nawaz 192325060 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty **Total Marks Awarded:**

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1) Evolution from standalone computing to cloud based

The evolution of computing has gone throught several stages.

* Standlone computing computer worked independently with local storage & software. Data sharing was limited.
* Client-server computing:- clients required services from centralized servers, Improving data sharing and management
* Distributed computing:- multiple computers worked together to complete tasks, Increasing performance and realibility
* grid computing:- resources from different locations were combined to solve large computional problems.
* cloud computing:- computing resources such as servers storage databases and applications are delivered over the Internet on demands.

Key - Technological Transition:-

* Development of high-speed Internet
* virtualization technology.
* Distribution and parallel computing
* Data centers with large scale Infrastructure.
* Service-oriented architecture
* Automation and resource management.

2) working of hypermissions is virtualization:-

A hypervision is a software that creates and manages multiple virtual machine (VMs) on a a single physical computer. It allocates cpu memory, storage, and network resources to each VM.

Features	type1 Hypervision	Type2 hypervision
Installation	Runs directly on hardware	Runs on a host operating system
performance	high	moderate.
security	more secure	less secure.
usage	Data centers cloud perforns	personal computers testing

Examples:-

Type 1:- VMware ESxi, microsoft hyper-V, xen

Type2:- oracle virtualbox, VMware workstation, parallel desktop.

3) Role of API'S In cloud service Integration:-

An API (Application programming Interfaces) enables communication between different software applications and cloud services.

Roles of APIs:-

* connect cloud applications and services.
* Enable data exchange between platform.
* Automate cloud operations.
* Allow developers to access cloud-resources programmatically.
* support application Integration across different cloud provides.

How APIs support Interoperability:-

* Use standard protocols such as HTTP and Rest.
* Exchange data in common formats like JSON and XML.
* Allow applications built on different technology to communicate.
* Enable Integration of SaaS, PaaS, and IaaS services

4) Resource polling in cloud computing:-

Resource polling is a cloud feature where computing resources (servers, storage, memory and networking) are shared among ~~memory~~ multiple users using a Multiple-tenant modes.

Benefits:-

* Efficient utilization of resources.
* Reduced operational cost
* Better scalability.
* High availability of services.
* Resources are allocated and reallocated based on user demands.

Illustration:-

A cloud provider maintains in a large pool of servers. Different organizations use the same infrastructure but each receives isolated virtual resources according to its requirements.

5) Rapid Elasticity in cloud computing:-

Rapid elasticity is the ability of cloud computing to quickly increase or decrease computing resources based on workload demands.

How it supports Dynamic workloads:-

* Automatically adds resources during high traffic
* Release used resources when demand decrease
* Ensure consistent application performance.

reduces costs by charging only for resources used.

* Supports suddens traffic spikes without service Interruption.

Example:-

An online shopping website experiences heavy traffic during a festival sale. The cloud automatically. Increases server. The cloud automatically Increases server capacity to handle more users. After the sale end their automatically cost reduces.